

EFFICIENT BAYESIAN INFERENCE FROM NOISY PAIRWISE COMPARISONS

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ABSTRACT

The evaluation of generative models remains a persistent methodological challenge, principally because conventional automatic metrics tend to correlate poorly with the perceptual and semantic judgments of human observers. Human assessment, though considerably more faithful to genuine preferences, is simultaneously costly and intrinsically noisy, insofar as participants differ appreciably in their expertise, attentiveness, and diligence. Pairwise comparison protocols mitigate a portion of this inconsistency, yet the subsequent aggregation of such comparisons into coherent overall quality scores necessitates careful probabilistic modeling. Bradley–Terry-based formulations, which iteratively update item scores from observed comparison outcomes, constitute the prevailing solution; existing instantiations, however, either disregard inter-rater variability altogether or forgo convergence guarantees, thereby compromising both robustness and interpretability. We introduce BBQ, a Bayesian Bradley–Terry variant that explicitly parameterizes rater quality—systematically downweighting, or altogether excluding, unreliable participants—and that furnishes a guarantee of monotonic likelihood convergence by virtue of an Expectation–Maximization procedure. Our empirical investigation demonstrates that, relative to baseline Bradley–Terry models, BBQ attains accelerated convergence, well-calibrated uncertainty estimates, and rankings that are at once more robust and more interpretable, even under noisy or crowdsourced annotation regimes. Taken together, this framework renders the human evaluation of generative models substantially more reliable and more cost-effective.

1 INTRODUCTION

The evaluation of generative models constitutes a formidable challenge, particularly in the case of large language models (LLMs) and image generators, for which standard automatic metrics frequently fail to capture the preferences of human observers. Metrics such as BLEU (Papineni et al., 2002) and perplexity (Jelinek, 1998) in the language domain, or PSNR (Gonzalez, 2009), MS-SSIM (Wang et al., 2003), and FID (Heusel et al., 2017) in the image domain, afford only a circumscribed view of perceived quality (Mentzer et al., 2020; CLIC, 2025; Chiang et al., 2024). It follows that human evaluation remains indispensable whenever meaningful rankings among competing models are to be established. Accordingly, leaderboards and comparative studies of large language models and Learned Image Compression (LIC) methods alike accord considerable weight to human preference data.

Human evaluation, however, is both expensive and time-consuming, which renders it imperative to devise protocols that are efficient while concurrently minimizing subjectivity and noise. Within this context, pairwise comparisons—in which participants elect between two items rather than assigning absolute scores—constitute an effective and eminently practical modality of human evaluation (Zerman et al., 2018; Wang et al., 2023). Not only are they generally less burdensome for participants, but they also yield judgments of markedly greater consistency. The exhaustive collection of all pairwise comparisons is nonetheless infeasible, since the number of distinct pairs grows quadratically with the number of items. Given only a limited subset of comparisons, elementary statistics such as raw win rates prove insufficient for deriving overall quality scores, precisely because a win rate is contingent upon the quality of the items against which an item happens to be compared.

To yield an overall ranking suitable for presentation on leaderboards, the individual pairwise comparisons must therefore be aggregated in a principled and effective manner—a requirement that motivates the development of aggregation methods that are at once robust and computationally efficient. The Bradley–Terry (BT) model (Bradley & Terry, 1952) addresses precisely this problem by iteratively updating item scores in accordance with observed comparison outcomes, and it has, in consequence, become the canonical approach to the aggregation of pairwise judgments across a broad range of domains.

Notwithstanding its demonstrated utility, the standard Bradley–Terry model is beset by several limitations of practical consequence. A variety of algorithms exists for estimating BT parameters, encompassing iterative scaling methods (Ford Jr, 1957), gradient descent, and the minorization–maximization (MM) algorithm (Hunter, 2004). Whereas the MM algorithm guarantees convergence to the global maximum for the elementary BT model, gradient-based methods—which bear a close affinity to the Elo rating system and enjoy widespread practical adoption—afford no comparable guarantee (Hunter, 2004). Moreover, for extended BT models that accommodate such factors as home-field advantage (Agresti, 2010), multiple comparisons (Plackett, 1975; Luce et al., 1959), or ties (Rao & Kupper, 1967), even MM algorithms may converge to nothing more than a local optimum (Hunter, 2004). A further and more fundamental difficulty is that human evaluation is inherently noisy: participants may rush, resort to guessing, or lapse in concentration, thereby engendering uneven reliability across the rater population. Should such variability be left unmodeled, unreliable raters are liable to distort item scores and to erode the stability of the resulting ranking.

We propose a Bayesian Bradley–Terry variant that jointly models item quality and rater reliability. To the best of our knowledge, no prior work has hitherto united the BT framework with a Bayesian formulation that explicitly parameterizes rater quality while simultaneously furnishing closed-form EM updates. Previous approaches that endeavor to account for noisy raters characteristically rely upon gradient-based optimization devoid of convergence guarantees; our method, by contrast, secures stable convergence and yields parameters amenable to direct interpretation. The Bayesian formulation, for its part, addresses epistemic uncertainty and confers a measure of regularization. In order to capture the variability inherent in participant behavior, we introduce rater-specific parameters that encode how consistent or trustworthy each participant is, thereby permitting the model to modulate the influence exerted by individual comparisons. From a latent-variable formulation we derive an EM algorithm that estimates item and rater parameters efficiently while guaranteeing monotonic improvement of the likelihood. This construction enables partially reliable raters to contribute meaningful information, even as it attenuates the influence of inattentive or inconsistent participants. By jointly modeling rater quality and adopting a Bayesian framework, our method additionally supplies uncertainty estimates for item scores, thereby facilitating rankings that are more interpretable and comparisons that remain robust across distinct studies.

We substantiate the effectiveness of our approach on a multiplicity of human evaluation datasets and, in addition, introduce a newly collected *inhomogeneous rater quality (IHQ)* dataset¹. Relative to standard BT and naive aggregation methods, our method exhibits accelerated convergence, enhanced robustness to noisy comparisons, and rankings of appreciably greater consistency. Beyond the evaluation of generative models per se, the proposed framework is applicable to a broad spectrum of scenarios in which noisy pairwise comparisons must be transmuted into reliable global rankings. In sum, our work advances the cause of more cost-efficient, interpretable, and reproducible human studies for the evaluation of AI-generated content.

2 RELATED WORK

Human studies frequently entail the ranking of items. Presenting participants with a choice between two items, rather than eliciting an absolute score for a single item, heightens sensitivity to subtle differences and concomitantly diminishes the variability of responses (Zerman et al., 2018; Wang et al., 2023). It is for this reason that a great many leaderboards for machine learning models are predicated upon pairwise comparisons. Under such a regime, participants declare their preference between two alternatives, whereupon these preferences are aggregated so as to yield an overall ranking. Prominent exemplars include the Chatbot Arena (Chiang et al., 2024) and the CLIC image

¹The IHQ dataset will be released upon acceptance.

compression challenge (CLIC, 2025), both of which elicit pairwise comparisons and combine them by means of variants of the Bradley–Terry (BT) model.

The Bradley–Terry model (Bradley & Terry, 1952), originally conceived for the ranking of competitors in sporting contests, furnishes a probabilistic framework within which to estimate the likelihood that one item is preferred over another. By converting pairwise comparisons into a coherent ranking, the model has established itself as a cornerstone of investigations spanning games, consumer preferences, and numerous cognate applications. Several extensions have subsequently been advanced with a view to broadening its applicability. The Plackett–Luce model, for instance, generalizes the BT framework from pairwise comparisons to rankings over multiple items (Plackett, 1975; Luce et al., 1959), defining a probability distribution over permutations through the multiplication of successive BT probabilities (Luce et al., 1959). Yet further modifications target specific contexts, among them the modeling of home-field advantage (Agresti, 2010), the incorporation of ties (Rao & Kupper, 1967), and the treatment of comparisons between groups rather than individuals (Huang et al., 2006).

Maximum likelihood estimation (MLE) is the method most commonly enlisted to infer the parameters of the basic BT model. Zermelo (1929) introduced an iterative procedure for computing these estimates, one that has since attained widespread adoption. Lange et al. (2000) subsequently demonstrated that this procedure constitutes a particular instance of minorization–maximization (MM) algorithms, which iteratively optimize surrogate functions so as to attain a local maximum of the likelihood. Hunter (2004) thereafter extended MM algorithms to generalized BT models and established the conditions under which convergence to the MLE is guaranteed. For the classical BT model the optimization is convex, whence convergence to the global optimum is assured (Hunter, 2004). For extensions such as home-field advantage, multiple comparisons, or ties, however, MM algorithms may converge to no more than a local maximum (Hunter, 2004).

Bayesian formulations of the BT model have likewise been explored as a means of incorporating prior knowledge and regularization (Adams, 2005; Guiver & Snelson, 2009; Caron & Doucet, 2012). Caron & Doucet (2012) established that MM algorithms admit interpretation as Expectation–Maximization (EM) procedures, thereby enabling Bayesian inference via Gibbs sampling. This formulation not only yields tractable complete-data likelihoods but also guarantees the convergence of the attendant Markov Chain Monte Carlo methods. The Bayesian perspective, moreover, smooths the posterior estimates—thus attenuating susceptibility to local maxima—and affords a principled quantification of uncertainty in the estimated item skills.

Annotator quality models constitute a critical extension, one that relaxes the untenable assumption that all comparisons are equally reliable. In crowdsourced evaluations, where participant quality may vary dramatically, this circumstance motivates approaches that explicitly accommodate annotator reliability (Chen et al., 2013). Chen et al. (2013) advanced a Bayesian model that jointly estimates item quality and rater reliability by means of annotator-specific parameters. Their formulation imposes Gaussian priors on both the item and rater parameters and introduces scaling factors within the likelihood in order to modulate the influence of individual annotators. Posterior inference is effected through gradient-based optimization, an approach that can prove problematic in high-dimensional spaces owing to the prospect of local optima; gradient-based methods, furthermore, offer no guarantee of convergence whatsoever. Our proposed EM-based method, by contrast, furnishes a guarantee of monotonic likelihood improvement. The Elo-based rater model developed by Google Research implements a simplified variant of this approach; it enjoys widespread use in the Challenge on Learned Image Compression (CLIC) (CLIC, 2025) and in related research (Mentzer et al., 2020; Ballé et al., 2025).

To the best of our knowledge, no prior work has combined the Bradley–Terry model with a Bayesian formulation that explicitly models rater quality while at the same time providing closed-form EM updates. In contradistinction to Chen et al. (2013), who resort to Gaussian priors and gradient-based optimization bereft of convergence guarantees, our approach capitalizes upon conjugate priors together with an EM algorithm that ensures monotonic likelihood improvement. And in contrast to the Elo-based rater model widely deployed in CLIC (CLIC, 2025)—a simplified heuristic predicated upon Elo-style updates—our method delivers a principled Bayesian treatment replete with uncertainty estimates and interpretable rater-quality parameters.

3 METHODOLOGY

3.1 BRADLEY-TERRY MODEL WITH RATER QUALITY

The task of converting a collection of noisy pairwise comparisons into a reliable ranking of items constitutes a fundamental problem in machine learning and statistics. Consider, then, a set of K items that are repeatedly compared with one another in pairs by a population of R raters; the data, which we denote by D , comprise the outcomes of these comparisons. For any two items i and j drawn from this set, Bradley & Terry (1952) proposed the following model:

$$P(i \text{ beats } j) = \frac{\lambda_i}{\lambda_i + \lambda_j} \quad (1)$$

where $\lambda_k > 0$ denotes a parameter associated with item $k \in \{1, 2, \dots, K\}$ that encodes its skill rating. This model affords a lucid and interpretable mechanism for inferring item scores from a corpus of observed wins and losses. It nonetheless rests upon the simplifying assumption that all comparisons are equally reliable—an assumption that, in the setting of human evaluation, is routinely violated. Participants may exhibit disparate degrees of expertise, attention, and diligence, and such heterogeneity gives rise to judgments that are unreliable and internally inconsistent.

To accommodate this variation in rater reliability, we introduce a rater-specific quality parameter $q_r \in [0, 1]$. Intuitively, with probability q_r rater r renders an informed judgment in accordance with the Bradley–Terry model, whereas with the complementary probability $1 - q_r$ the rater guesses at random, as though flipping a fair coin between the two items. This intuition gives rise to the following mixture model for the probability that rater r ranks item i above item j :

$$P(r \text{ ranks } i \text{ above } j) = q_r \left(\frac{\lambda_i}{\lambda_i + \lambda_j} \right) + (1 - q_r) \left(\frac{1}{2} \right) \quad (2)$$

The log-likelihood of the data is given by:

$$\begin{aligned} \log P(D \mid \lambda, q) = & \sum_{r=1}^R \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K \left[w_{r,ij} \log \left(q_r \frac{\lambda_i}{\lambda_i + \lambda_j} + (1 - q_r) \frac{1}{2} \right) \right. \\ & \left. + (n_{r,ij} - w_{r,ij}) \log \left(q_r \frac{\lambda_j}{\lambda_i + \lambda_j} + (1 - q_r) \frac{1}{2} \right) \right], \end{aligned} \quad (3)$$

where $n_{r,ij}$ denotes the total number of comparisons between items i and j performed by rater r , and $w_{r,ij}$ the number of occasions on which rater r ranks item i above item j . This likelihood admits no closed-form maximum likelihood estimator, and consequently the optimal parameters λ and q cannot be derived analytically. Although an iterative optimization method such as gradient descent might in principle be employed, it affords no guarantee of convergence. The Expectation–Maximization (EM) algorithm, by contrast, obviates the tuning of a learning rate and guarantees both a monotonic increase of the observed-data likelihood and convergence to a stationary point (Dempster et al., 1977).

3.2 THURSTONIAN INTERPRETATION

Following Caron & Doucet (2012), we interpret the Bradley–Terry model within a Thurstonian framework (Diaconis, 1988). From this vantage point, a comparison between items i and j is conceptualized as a race in which each item is endowed with a random arrival time, Y_i and Y_j respectively. These arrival times are assumed to follow exponential distributions:

$$Y_i \sim \mathcal{E}(\lambda_i), \quad Y_j \sim \mathcal{E}(\lambda_j), \quad (4)$$

and the item exhibiting the smaller arrival time is declared the winner. This construction leads directly to the standard Bradley–Terry probability:

$$P(i \text{ beats } j) = P(Y_i < Y_j) = \frac{\lambda_i}{\lambda_i + \lambda_j}. \quad (5)$$

For the purposes of the EM algorithm, we introduce latent variables so as to simplify the complete-data likelihood. We first define an indicator variable

$$A_{r,ij}^{(c)} \sim \text{Bernoulli}(q_r), \quad (6)$$

which registers whether the c -th comparison of items i and j by rater r conforms to the Bradley–Terry model. By means of these indicators, we define the latent variable $Z_{r,ij}$ as the sum of the minimal arrival times across the $n_{r,ij}$ comparisons performed by rater r :

$$Z_{r,ij} = \sum_{c=1}^{n_{r,ij}} A_{r,ij}^{(c)} \min(Y_i^{(c)}, Y_j^{(c)}). \quad (7)$$

Conditioned on $m_{r,ij} = \sum_{c=1}^{n_{r,ij}} A_{r,ij}^{(c)}$, the variable $Z_{r,ij}$ follows a Gamma distribution,

$$Z_{r,ij} \mid m_{r,ij} \sim \Gamma(m_{r,ij}, \lambda_i + \lambda_j), \quad (8)$$

where $\Gamma(\alpha, \beta)$ denotes the Gamma distribution parameterized by shape α and inverse scale β . This Gamma-distributed latent-variable formulation renders the EM update tractable while nonetheless accounting for rater-specific quality.

3.3 EXPECTATION-MAXIMIZATION UPDATES

The EM algorithm is an iterative procedure for obtaining maximum a posteriori (MAP) estimates of our model parameters, λ and q , and it accomplishes this by treating the rater quality together with the unobserved arrival times of the Thurstonian interpretation as latent variables. The procedure is, moreover, guaranteed to converge to a stationary point of the posterior distribution.

We begin by specifying prior distributions over the parameters. The item skills λ_k are assigned a Gamma prior, $\lambda_k \sim \Gamma(a, b)$, which is conjugate to the exponential distribution, while each rater’s quality parameter q_r is endowed with a Beta prior, $q_r \sim B(\alpha, \beta)$.

At the heart of the EM algorithm lies the iterative maximization of the expected complete-data log-posterior, conditioned upon the current parameter estimates (λ^*, q^*) . The algorithm proceeds by alternating between the following two steps until convergence is achieved:

E-step: Expectation In the E-step we compute the expected complete-data log-posterior, a function we denote by Q . This function represents the expectation of the log-posterior over all observed and latent variables, taken conditionally on the observed data and on the current parameter estimates inherited from the preceding iteration. It is defined as follows:

$$Q(\lambda, q \mid \lambda^*, q^*) = \mathbb{E}_{A, Z \mid D, \lambda^*, q^*} [\ell_c(\lambda, q; D, Z, A) + \log P(\lambda) + \log P(q)]. \quad (9)$$

The complete-data log-likelihood, ℓ_c , may in turn be decomposed into three constituent components:

$$\ell_c(\lambda, q; D, Z, A) = \log P(Z \mid D, A, \lambda, q) + \log P(A \mid D, \lambda, q) + \log P(D \mid \lambda, q). \quad (10)$$

M-step: Maximization This step updates the model parameters by maximizing the Q function. By capitalizing upon the expectations computed in the E-step, the M-step transmutes the original, intractable optimization problem into a set of simpler closed-form updates. The pivotal quantity entering both updates is the posterior probability that a given comparison rendered by rater r conforms to the Bradley–Terry model. This quantity, which we denote by γ , represents the weight accorded to a rater’s judgment as a function of the degree to which it accords with the model’s current predictions. It is given by:

$$\gamma_{r,ij}^{(t-1)} = \frac{q_r^{(t-1)} y_{ij}^{(t-1)}}{q_r^{(t-1)} y_{ij}^{(t-1)} + (1 - q_r^{(t-1)}) \frac{1}{2}}, \quad (11)$$

where $y_{ij}^{(t-1)} = \frac{\lambda_i^{(t-1)}}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}}$ is the Bradley–Terry probability that item i beats item j . This posterior probability γ encodes our confidence that a given comparison was meaningful rather than random, conditioned upon the current parameter estimates; larger values of γ correspond, accordingly, to more trustworthy comparisons.

The updated estimate of a rater’s quality, q_r , takes the form of a weighted average. The numerator accumulates the “effective number of wins” attributable to that rater, wherein each win is weighted by the posterior probability (γ) that it constituted a meaningful, non-random judgment; this sum is then combined with the hyperparameters of the Beta prior so as to regularize the estimate. The denominator, in turn, normalizes the sum by the total number of comparisons together with the prior parameters. Intuitively, the update elevates a rater’s quality score whenever that rater’s judgments accord frequently with the model’s predictions. The update is given by:

$$q_r^{(t)} = \frac{\sum_{i=1}^K \sum_{j=i+1}^K \left[w_{r,ij} \gamma_{r,ij}^{(t-1)} + w_{r,ji} \gamma_{r,ji}^{(t-1)} \right] + (\alpha - 1)}{n_r + \alpha + \beta - 2}, \quad (12)$$

where n_r is the total number of comparisons by rater r .

The updated estimate of an item’s skill, λ_i , is a ratio that holds two key quantities in balance. The numerator aggregates the “effective wins” of item i across all raters, with each win once again weighted by the rater’s quality (γ); this term thus embodies the total positive evidence accrued in favor of item i . The denominator, by contrast, accounts for the totality of comparisons in which item i participated and functions as a normalizing factor. Both terms are further regularized by the hyperparameters of the Gamma prior. The update is given by:

$$\lambda_i^{(t)} = \frac{\sum_{r=1}^R \left[\sum_{j=1, j \neq i}^K w_{r,ij} \gamma_{r,ij}^{(t-1)} \right] + (a - 1)}{\sum_{j=1, j \neq i}^K \left[\frac{\sum_{r=1}^R \left[w_{r,ij} \gamma_{r,ij}^{(t-1)} + w_{r,ji} \gamma_{r,ji}^{(t-1)} \right]}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}} \right] + b}. \quad (13)$$

A complete derivation of the Expectation–Maximization algorithm is furnished in Appendix B.

4 EXPERIMENTAL RESULTS

In order to assess the performance of our **Bayesian Bradley–Terry** model with rater **Quality (BBQ)**, we undertake a series of experiments in which it is pitted against two baselines: the Bayesian Bradley–Terry model (Bayes-BT) (Caron & Doucet, 2012) and a gradient-descent-based BT model that incorporates rater quality (Crowd-BT) (Chen et al., 2013). For Crowd-BT we adopt the implementation furnished by Google Research (Google, 2025), the very implementation employed by both CLIC (2025) and Ballé et al. (2025). The hyperparameters governing our experiments are documented in Section C.

We evaluate the performance of several Bradley–Terry (BT) variants across a diversity of datasets, encompassing human preference benchmarks for large language models (HUMAINE (Team, 2025), MT-Bench (Zheng et al., 2023)) and for image compression (WD (Ballé et al., 2025), HiFiC (Mentzer et al., 2020), ConHa (Aczel & Wattenhofer, 2024)), together with a newly collected inhomogeneous rater quality (IHQ) dataset derived from the CLIC2024 image test set (CLIC, 2025). So as to isolate the effect of rater quality, we partition the IHQ dataset into two subsets, screened and unscreened, a stratification that permits systematic analysis of the manner in which low-quality, high-quality, and mixed comparisons bear upon model performance. The particulars of the IHQ dataset’s construction are set out in Section E, and a summary of all datasets—including the counts of comparisons, raters, and items—is provided in Section D and Table 2.

Ground-truth rankings are unavailable for the real datasets. We therefore first validated the models on simulated datasets, on which all methods recover an identical ordering as the number of samples grows. For the crowd-sourced datasets, we approximate the ground truth by the ranking obtained on the dataset in its entirety. That this constitutes a reasonable approximation of the true ordering for real-world datasets we corroborate by inspecting the top-1 item within each ranking. For every dataset save IHQ-unscreened, all three models recover the same top-ranked item. On the IHQ-unscreened dataset, however, Crowd-BT and Bayes-BT both fail to identify the reference image as the highest-quality item, whereas BBQ succeeds in doing so.

A reliable aggregation method ought to reproduce the same ordering were the study to be repeated. We approximate this notion of stability by means of bootstrapping, which yields an estimate of the variability inherent in the rankings; the attendant details are described in Section F. It bears noting

that the WD and HiFiC studies both employed active selection of comparison pairs—HiFiC by way of a binary-search strategy, WD by selecting pairs so as to maximize information gain. For this reason, the bootstrapping results on these two datasets ought to be construed with a measure of caution.

We assess performance by recourse to two complementary metrics. Top-1 agreement designates the fraction of bootstrapped samples that identify the same best item as does the full dataset. Kendall’s Tau (τ), for its part, measures the ordinal correlation between rankings (Kendall, 1938), with larger values betokening greater stability. Top-1 agreement is of paramount relevance when it is the single best model that matters, whereas Kendall’s Tau gauges the stability of the ranking taken as a whole.

4.1 PERFORMANCE ACROSS DATASETS

We compare the models across datasets in order to appraise their performance with respect to both ranking accuracy and stability. Top-1 agreement and Kendall’s Tau are summarized in Table 1. Datasets furnishing a greater number of comparisons per model—among them MT-Bench, WD, and HiFiC—generally prove more tractable. HUMAINE, intriguingly, departs from this trend, thereby underscoring that factors beyond the sheer volume of comparisons, such as the consistency and diversity of the raters, can materially influence model performance.

Taken in the aggregate, BBQ exhibits superior stability and robustness across datasets. It identifies the top-performing item with the greatest frequency, ranking as the shared best on three datasets, and it recovers the overall ranking most accurately on five of the eight datasets, placing second on the three that remain.

The three datasets on which BBQ ranks second in Kendall’s Tau are precisely those comprising high-quality, homogeneous rater sets. On MT-Bench, all models recover the top item flawlessly. The WD dataset, collected by Ballé et al. (2025), comprises a mere five raters—in all likelihood the paper’s own authors—a circumstance indicative of exceptionally scrupulous evaluation. In IHQ-screened, the raters were explicitly filtered for quality. In such settings, where rater quality is uniformly high, the explicit modeling of rater reliability confers scant advantage, and the margin of BBQ over simpler models is correspondingly diminished.

Experiments on the IHQ dataset, considered across both its screened and unscreened rater subsets, disclose an unmistakable pattern. BBQ outperforms Crowd-BT and Bayes-BT by a substantial margin whenever low-quality raters are present, as is the case for CLIC-all and CLIC-unscreened. All three models attain their highest Top-1 accuracy when confined to the screened subset, a finding that throws the importance of rater selection into sharp relief. Although Crowd-BT does explicitly model rater quality, its performance deteriorates appreciably on the full dataset, presumably because the crowdsourced raters each contribute fewer than 40 comparisons, which heightens vulnerability to noisy annotations. BBQ, by contrast, sustains strong performance even in the absence of screening, suffering only a marginal decrement in Top-1 accuracy—a testament to its robustness in the face of low-quality raters.

The attainment of uniformly high rater quality is, nevertheless, a considerable challenge. Large-scale crowdsourcing inevitably introduces variability, screening procedures exact a substantial cost, and subjective factors may impinge upon even the most diligent of annotators. Against this backdrop, BBQ furnishes a principled means of harnessing the information supplied by partially reliable raters while simultaneously curtailing the impact of noisy contributions.

4.2 SCALING WITH RATERS AND COMPARISONS

As was observed in Table 1, datasets affording more comparisons per model generally engender superior performance. The number of comparisons may be augmented along either of two axes: by enlisting additional raters, or by increasing the number of comparisons each rater performs.

Figure 1 illustrates the influence of both factors upon the three BT variants. The left column depicts performance as a function of the number of raters (with the number of comparisons per rater held fixed at its maximum), whereas the right column depicts performance as a function of the number of comparisons per rater (with the number of raters held fixed at its maximum). A pronounced disparity in performance manifests itself in Top-1 accuracy, even as Kendall’s τ remains largely

Table 1: Performance of the several BT-based aggregation methods across a range of datasets. Top: Top-1 agreement [%]; Bottom: Kendall’s Tau. BBQ identifies the top-performing item more frequently than any competitor across all datasets, recovers the best overall ranking on more than half of them, and places second on those that remain.

	HUMAINE	MT-Bench	WD	HiFiC	ConHa	IHQ		
						all	scr.	unscr.
Top-1 Agreement [%]								
Crowd-BT	85.60	100.00	99.29	98.63	<u>66.59</u>	<u>85.31</u>	98.57	<u>33.15</u>
Bayes-BT	<u>97.30</u>	100.00	100.00	100.00	57.30	75.07	<u>98.90</u>	24.32
BBQ (ours)	97.50	100.00	100.00	100.00	77.12	99.32	99.80	61.92
Kendall’s Tau								
Crowd-BT	0.9385	0.9743	0.9279	<u>0.9366</u>	0.9180	<u>0.9245</u>	0.9171	0.8482
Bayes-BT	<u>0.9443</u>	0.9569	0.9459	0.9293	0.9110	0.9240	0.9116	<u>0.8507</u>
BBQ (ours)	0.9463	<u>0.9675</u>	<u>0.9359</u>	0.9525	0.9265	0.9270	<u>0.9132</u>	0.8563

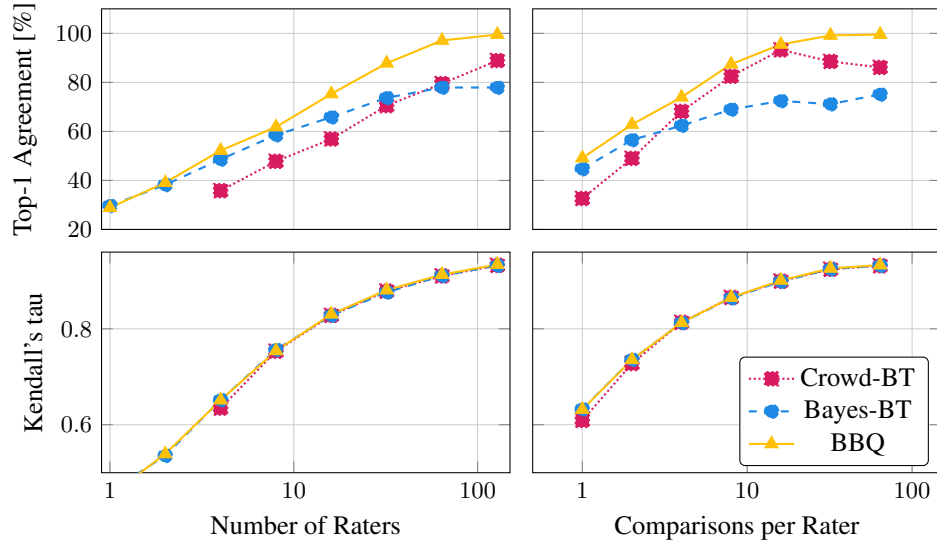


Figure 1: Scaling behavior of the Bradley–Terry variants (Crowd-BT (Caron & Doucet, 2012), Bayes-BT (Chen et al., 2013), BBQ (ours)) on the IHQ-all dataset. **Left:** performance as a function of the number of raters. **Right:** performance as a function of the number of comparisons per rater. Both Top-1 agreement and Kendall’s τ improve appreciably as raters or comparisons are added. Whereas Top-1 agreement discriminates among the models, Kendall’s τ remains largely comparable across them. Crowd-BT fails to converge when raters are exceedingly few, thereby underscoring the advantage conferred by the EM algorithm. Crowd-BT and BBQ perform comparably under sparse data, yet BBQ surpasses Bayes-BT as the number of raters or comparisons grows.

uniform across models. Crowd-BT fails to converge when confronted with only one or two raters, a shortcoming that accentuates the advantage of the EM algorithm, which—unlike gradient descent—affords convergence guarantees.

Crowd-BT and BBQ perform comparably when the number of raters or comparisons per rater is small. BBQ requires both a plurality of raters and a plurality of comparisons per rater in order to discriminate effectively among rater qualities. As either quantity increases, BBQ comes to outperform Bayes-BT by an ever-widening margin. Bayes-BT, by contrast, tends to underperform overall, yet it can nonetheless surpass Crowd-BT when data per rater are scarce or raters are few, precisely because Crowd-BT is unable to estimate rater quality reliably in such sparse regimes.

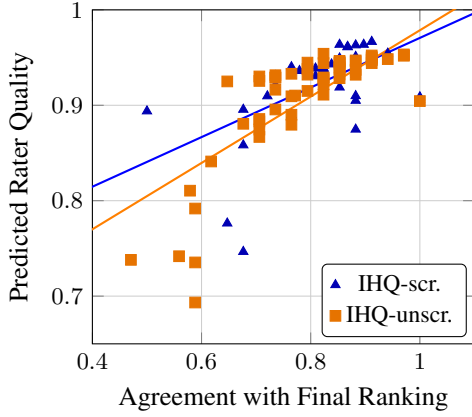


Figure 2: Scatter plot relating each rater’s agreement with the final ranking (horizontal axis) to the rater quality predicted by the model (vertical axis) for the IHQ datasets. Every point corresponds to an individual rater; triangles denote the filtered dataset, while squares denote the unfiltered dataset.

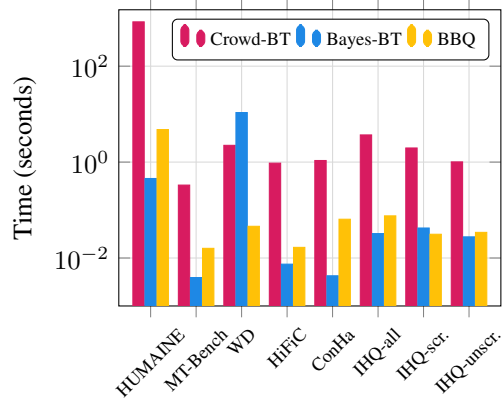


Figure 3: Average computation time in seconds (logarithmic scale) for the three models, measured on a single bootstrapped sample across eight datasets. Whereas Crowd-BT may demand substantial computation on certain datasets, BBQ remains uniformly fast throughout.

4.3 RATER QUALITY

Figure 2 depicts the relationship between agreement with the final ranking and the predicted rater quality. We discern a distinct positive correlation: raters whose judgments align more closely with the consensus ranking are accorded higher quality by the model. For the filtered dataset, the Pearson correlation stands at $r = 0.551$ across 50 raters, whereas for the unfiltered dataset it is stronger still, at $r = 0.724$ across 62 raters.

As one would anticipate, the unscreened dataset harbors several raters exhibiting both diminished agreement and diminished predicted quality. Certain raters in the IHQ-unscreened dataset even descend below the threshold of random guessing (50% agreement), dissenting systematically from the majority. This observation lays bare the importance of modeling rater quality when pairwise comparison data are aggregated. BBQ successfully detects such raters and assigns them commensurately lower quality scores, thereby curtailing their influence upon the final ranking and mitigating the noise they would otherwise introduce.

4.4 COMPUTATIONAL EFFICIENCY

Figure 3 reports the average computation time, in seconds, that each method requires to process a single bootstrapped sample across the various datasets. These timings afford a pragmatic perspective on the feasibility of the methods in real-world evaluation scenarios. BBQ converges within a few seconds on every dataset without exception. Crowd-BT, by comparison, demands more time than BBQ across the board, incurring particularly protracted runtimes on datasets with a great many comparisons, such as HUMANE, where convergence consumes on the order of 15 minutes. Bayes-BT converges sluggishly on the WD dataset, though on the remaining datasets it proves marginally faster than BBQ. The efficiency of BBQ is attributable to the closed-form EM updates derived within our method, which enable rapid convergence even on sizeable datasets.

It merits emphasis, moreover, that Crowd-BT was highly optimized and implemented in C (Kernighan & Ritchie, 1988), whereas BBQ was implemented in plain NumPy without any dedicated optimization. This disparity intimates that BBQ could be rendered faster still by way of a compiled or vectorized implementation. It follows that BBQ is not merely more robust and stable but also eminently practical for large-scale human preference studies and for applications predicated upon repeated bootstrapping. The confluence of accuracy, stability, and speed renders BBQ a compelling choice for deployment in real-world settings.

4.5 CONFIDENCE INTERVALS

We now examine how Crowd-BT, Bayes-BT, and BBQ estimate the uncertainty attaching to item rankings. Within a controlled simulation, raters furnish pairwise comparisons of items of equal skill (H_0). Each model estimates the skills and constructs uncertainty intervals with a view to testing for significant differences (Section G). The resulting Type I error rates are displayed in Figure 6. At the 99% confidence level the expected error rate is 1%. Crowd-BT and BBQ align closely with this expectation, yielding rates slightly below it, whereas Bayes-BT proves more conservative, hovering around $\sim 0.1\%$. Crowd-BT and BBQ thus deliver well-calibrated uncertainty estimates, while Bayes-BT systematically underestimates the incidence of false positives.

5 LIMITATIONS

Although BBQ models rater quality to good effect, its advantages recede in settings in which all raters are uniformly reliable. In such homogeneous datasets, the explicit modeling of variability yields little additional benefit. Guaranteeing consistently high-quality raters, however, frequently exacts considerable cost and effort, a burden that may prove infeasible in large-scale studies. This tension makes plain that BBQ is of greatest utility in realistic crowdsourced settings, and of correspondingly lesser utility when evaluations have been carefully curated.

A further limitation pertains to the class of data the framework is equipped to handle. BBQ is at present confined to pairwise comparisons, whereas many human evaluation studies elicit ratings, rankings, or multi-way inputs. Extending the model to accommodate these alternative forms of feedback would broaden its applicability considerably.

From a methodological standpoint, our experiments rest upon a limited number of bootstrapped comparisons. While this affords a practical gauge of stability, larger-scale studies would be requisite to validate robustness more conclusively. The model, furthermore, presumes independence across comparisons and makes no allowance for the contextual or order effects that may sway human judgments in practice.

Finally, although BBQ scales gracefully in computational terms, exceedingly large numbers of items or raters could nonetheless present difficulties in the absence of optimized or compiled implementations.

6 CONCLUSION

We have introduced BBQ, a Bayesian Bradley–Terry model that jointly estimates item quality and rater reliability. Our central contribution is the derivation of an Expectation–Maximization (EM) algorithm that estimates item skills and individual rater quality in concert, thereby down-weighting or altogether excising the influence of unreliable participants. By modeling rater quality explicitly, the method yields rankings that are at once more stable and more accurate. The EM algorithm, moreover, guarantees rapid convergence together with monotonic likelihood improvement, and in so doing redresses the limitations that beset gradient-based approaches.

Across a diverse array of datasets, BBQ consistently attained high Top-1 agreement together with superior Kendall’s Tau, evincing both robustness and reliability. The model excels in scenarios characterized by noisy or crowdsourced raters, preserving its accuracy even when individual raters contribute but few comparisons. BBQ proves especially effective in large-scale settings in which many raters are non-experts or diverge widely in attentiveness and diligence. Our results underscore that the incorporation of rater quality is of particular consequence whenever evaluation quality is heterogeneous or only partially reliable. The Bayesian framework, in addition, supplies principled uncertainty estimates for item scores, thereby enabling interpretable comparisons across distinct studies. We further demonstrate that the model’s error bars are well-calibrated and may be enlisted to adjudicate whether the differences between items are statistically significant. The predicted rater quality, finally, coheres with agreement to the final rankings, thus corroborating the model’s capacity to discern reliable evaluators.

In sum, BBQ advances the methodology of human evaluation by furnishing a practical, interpretable, and generalizable approach to the aggregation of noisy pairwise comparisons. This work thereby

540 constitutes a significant stride toward more cost-effective, interpretable, and reproducible human
541 studies for the evaluation of AI-generated content.
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A USAGE OF LLMs

In the course of preparing this paper, we availed ourselves of large language models (LLMs) as supportive tools. ChatGPT, Claude, Gemini, and Grammarly assisted with spellchecking, the refinement of phrasing, and the condensation of text so as to enhance clarity and readability. ChatGPT, Claude, and Cursor were additionally enlisted for code analysis, code completion, and the generation of visualizations in aid of our development workflow. These tools were employed solely in an auxiliary capacity, whereas all core research ideas, the experimental design, and the interpretation of results were developed independently by the authors.

B BAYESIAN BT WITH RATER QUALITY DERIVATION

We consider R raters comparing K items. Each rater r is endowed with a quality q_r , in the sense that with probability q_r the rater follows the Bradley–Terry model, and with the complementary probability $1 - q_r$ chooses at random. What follows is a standard EM (Dempster et al., 1977) derivation that incorporates these rater-specific reliabilities into the Bayesian estimation framework.

B.1 MODEL DEFINITION

Notation:

- $\lambda = (\lambda_1, \dots, \lambda_K)$: skill parameters for the K items.
- $q = (q_1, \dots, q_R)$: quality parameters for the R raters.
- $Y_i \sim \mathcal{E}(\lambda_i)$: latent arrival time associated with item i .
- $A_{r,ij}^{(c)} \sim \text{Bernoulli}(q_r)$: indicator that the c -th comparison of pair (i, j) by rater r follows the Bradley-Terry model.
- $n_{r,ij}$: total number of comparisons of (i, j) by rater r .
- $n_{ij} = \sum_{r=1}^R n_{r,ij}$: total number of comparisons of (i, j) .
- $w_{r,ij}$: number of times rater r ranked i above j .
- $w_{ij} = \sum_{r=1}^R w_{r,ij}$: total number of times i was ranked above j .
- $m_{r,ij} = \sum_{c=1}^{n_{r,ij}} A_{r,ij}^{(c)}$: number of Bradley-Terry-model comparisons of (i, j) by rater r .
- $m_{ij} = \sum_{r=1}^R m_{r,ij}$: total number of Bradley-Terry-model comparisons of (i, j) .
- $v_{r,ij} = \sum_{c=1}^{w_{r,ij}} A_{r,ij}^{(c)}$: number of Bradley-Terry-model wins of i over j by rater r .
- $v_{ij} = \sum_{r=1}^R v_{r,ij}$: total number of Bradley-Terry-model wins of i over j .
- $Z_{r,ij} = \sum_{c=1}^{n_{r,ij}} A_{r,ij}^{(c)} \min(Y_i^{(c)}, Y_j^{(c)})$: sum of minimal arrival times, with

$$Z_{r,ij} \mid m_{r,ij} \sim \Gamma(m_{r,ij}, \lambda_i + \lambda_j).$$

Bradley-Terry probability:

$$P(i \text{ beats } j) = \frac{\lambda_i}{\lambda_i + \lambda_j}.$$

Mixture with rater quality:

$$P(r \text{ ranks } i \text{ above } j) = q_r \frac{\lambda_i}{\lambda_i + \lambda_j} + (1 - q_r) \frac{1}{2}.$$

Latent exponential view:

$$Y_i \sim \mathcal{E}(\lambda_i), \quad Y_j \sim \mathcal{E}(\lambda_j), \quad P(i \text{ beats } j) = P(Y_i < Y_j).$$

B.2 COMPLETE-DATA LOG-LIKELIHOOD

We need to compute:

$$\begin{aligned}\ell_c(\lambda, q; D, Z, A) &= \log P(D, Z, A \mid \lambda, q) \\ &= \log P(Z \mid D, A, \lambda, q) + \log P(A \mid D, \lambda, q) + \log P(D \mid \lambda, q)\end{aligned}$$

Log-Likelihood of $P(Z \mid A, \lambda)$:

$$\begin{aligned}\log P(Z \mid D, A, \lambda, q) &= \sum_{i=1}^K \sum_{j=i+1}^K \left[m_{ij} \log(\lambda_i + \lambda_j) - (\lambda_i + \lambda_j) z_{ij} \right. \\ &\quad \left. + (m_{ij} - 1) \log z_{ij} - \log \Gamma(m_{ij}) \right]\end{aligned}$$

Log-Likelihood of $P(A \mid q)$:

$$\log P(A \mid D, \lambda, q) = \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[m_{r,ij} \log q_r + (n_{r,ij} - m_{r,ij}) \log(1 - q_r) \right]$$

Log-Likelihood of $P(D \mid \lambda, A)$:

$$\begin{aligned}\log P(D \mid \lambda, q) &= \sum_{r=1}^R \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K \left[v_{r,ij} \log \frac{\lambda_i}{\lambda_i + \lambda_j} + (w_{r,ij} - v_{r,ij}) \log \frac{1}{2} \right] \\ &= \sum_{r=1}^R \sum_{i=1}^K [v_{ir} \log \lambda_i] \\ &\quad - \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K [(v_{r,ij} + v_{r,ji}) \log(\lambda_i + \lambda_j) + (w_{r,ij} + w_{r,ji} - v_{r,ij} - v_{r,ji}) \log 2] \\ &= \sum_{i=1}^K [v_i \log \lambda_i] \\ &\quad - \sum_{i=1}^K \sum_{j=i+1}^K [m_{ij} \log(\lambda_i + \lambda_j) + (n_{ij} - m_{ij}) \log 2]\end{aligned}$$

Complete-Data Log-Likelihood:

$$\begin{aligned}
\ell_c(\lambda, q; D, Z, A) &= \log P(Z \mid D, A, \lambda, q) + \log P(A \mid D, \lambda, q) + \log P(D \mid \lambda, q) \\
&= \sum_{i=1}^K \sum_{j=i+1}^K \left[m_{ij} \log(\lambda_i + \lambda_j) - (\lambda_i + \lambda_j) z_{ij} + (m_{ij} - 1) \log z_{ij} - \log \Gamma(m_{ij}) \right] \\
&\quad + \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[m_{r,ij} \log q_r + (n_{r,ij} - m_{r,ij}) \log(1 - q_r) \right] \\
&\quad + \sum_{i=1}^K [v_i \log \lambda_i] \\
&\quad - \sum_{i=1}^K \sum_{j=i+1}^K [m_{ij} \log(\lambda_i + \lambda_j) + (n_{ij} - m_{ij}) \log 2] \\
&= \sum_{i=1}^K \sum_{j=i+1}^K \left[(n_{ij} - m_{ij}) \log 2 - (\lambda_i + \lambda_j) z_{ij} + (m_{ij} - 1) \log z_{ij} - \log \Gamma(m_{ij}) \right] \\
&\quad + \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[m_{r,ij} \log q_r + (n_{r,ij} - m_{r,ij}) \log(1 - q_r) \right] \\
&\quad + \sum_{i=1}^K [v_i \log \lambda_i]
\end{aligned}$$

B.3 EXPECTATION STEP

We introduce conjugate priors: Gamma distribution $\lambda_i \sim \Gamma(a, b)$ for each item i , and Beta $q_r \sim B(\alpha, \beta)$ for each rater r .

The Q -function is the expectation of the complete-data log-posterior:

$$Q(\lambda, q \mid \lambda^*, q^*) = \mathbb{E}_{A, Z \mid D, \lambda^*, q^*} [\ell_c(\lambda, q; D, Z, A) + \log P(\lambda) + \log P(q)].$$

Posterior probability of a Bradley-Terry-consistent annotation:

$$\begin{aligned}
P(A_{r,ij}^{(k)} = 1 \mid i \succ j) &= \frac{P(i \succ j \mid A = 1) P(A = 1)}{P(i \succ j \mid A = 1) P(A = 1) + P(i \succ j \mid A = 0) P(A = 0)} \\
&= \frac{\left(\frac{\lambda_i^*}{\lambda_i^* + \lambda_j^*} \right) q_r^*}{\left(\frac{\lambda_i^*}{\lambda_i^* + \lambda_j^*} \right) q_r^* + \frac{1}{2} (1 - q_r^*)} = \gamma_{r,ij}^*.
\end{aligned}$$

Expected sufficient statistics:

$$\begin{aligned}
\mathbb{E}[m_{r,ij}] &= w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^* \\
\mathbb{E}[m_{ij}] &= \sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) \\
\mathbb{E}[v_{r,ij}] &= w_{r,ij} \gamma_{r,ij}^* \\
\mathbb{E}[v_{ir}] &= \sum_{j: j \neq i}^K w_{r,ij} \gamma_{r,ij}^* \\
\mathbb{E}[v_i] &= \sum_{r=1}^R \sum_{j: j \neq i}^K w_{r,ij} \gamma_{r,ij}^* \\
\mathbb{E}[z_{ij}] &= \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*}.
\end{aligned}$$

Expected complete-data log-posterior:

$$\begin{aligned}\mathbb{E}[\ell_c] = & \sum_{i=1}^K \sum_{j=i+1}^K \left[-(\lambda_i + \lambda_j) \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} \right] \\ & + \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[(w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) \log q_r \right. \\ & \quad \left. + (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) \log(1 - q_r) \right] \\ & + \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K w_{r,ij} \gamma_{r,ij}^* \log \lambda_i \\ & + \text{const.}\end{aligned}$$

Priors contribute:

$$\begin{aligned}\mathbb{E}[\log P(\lambda)] &= \sum_{i=1}^K [(a-1) \log \lambda_i - b \lambda_i], \\ \mathbb{E}[\log P(q)] &= \sum_{r=1}^R [(\alpha-1) \log q_r + (\beta-1) \log(1 - q_r)].\end{aligned}$$

Final Q -function:

$$\begin{aligned}Q(\lambda, q \mid \lambda^*, q^*) = & \sum_{i=1}^K \sum_{j=i+1}^K \left[-(\lambda_i + \lambda_j) \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} \right] \\ & + \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[(w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) \log q_r \right. \\ & \quad \left. + (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) \log(1 - q_r) \right] \\ & + \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K w_{r,ij} \gamma_{r,ij}^* \log \lambda_i \\ & + \sum_{i=1}^K [(a-1) \log \lambda_i - b \lambda_i] \\ & + \sum_{r=1}^R [(\alpha-1) \log q_r + (\beta-1) \log(1 - q_r)] \\ & + \text{const.}\end{aligned}$$

where

$$\gamma_{r,ij}^* = \frac{q_r^* \frac{\lambda_i^*}{\lambda_i^* + \lambda_j^*}}{q_r^* \frac{\lambda_i^*}{\lambda_i^* + \lambda_j^*} + (1 - q_r^*) \frac{1}{2}}.$$

B.4 M-STEP

The M-step maximizes the Q -function with respect to the parameters (λ, q) , holding fixed the expectations computed in the E-step.

Update for q_r The update for each rater quality q_r is obtained by maximizing Q with respect to q_r , inclusive of the contribution of the Beta prior.

$$\begin{aligned}
Q(q \mid \lambda^*, q^*) &= \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[(w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) \log q_r \right. \\
&\quad \left. + (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) \log(1 - q_r) \right] \\
&\quad + (\alpha - 1) \log q_r + (\beta - 1) \log(1 - q_r) + \text{const.} \\
\frac{\partial Q}{\partial q_r} &= \sum_{i=1}^K \sum_{j=i+1}^K \left[\frac{w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*}{q_r} - \frac{n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{1 - q_r} \right] \\
&\quad + \frac{\alpha - 1}{q_r} - \frac{\beta - 1}{1 - q_r} \\
&\stackrel{0}{=} \frac{\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) + (\alpha - 1)}{q_r} \\
&\quad - \frac{\sum_{i=1}^K \sum_{j=i+1}^K (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) + (\beta - 1)}{1 - q_r} \\
&\Rightarrow (1 - q_r) \left(\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) + (\alpha - 1) \right) \\
&\Rightarrow q_r \left(\sum_{i=1}^K \sum_{j=i+1}^K (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) + (\beta - 1) \right) \\
&\Rightarrow q_r = \frac{\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) + (\alpha - 1)}{\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^* + n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) + (\beta + \alpha - 2)} \\
&\Rightarrow q_r^{(t)} = \frac{\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^{(t-1)} + w_{r,ji} \gamma_{r,ji}^{(t-1)}) + (\alpha - 1)}{n_r + \beta + \alpha - 2}
\end{aligned}$$

where the last line furnishes the explicit update at iteration t .

Update for λ_i The update for each item skill λ_i is obtained by maximizing Q with respect to λ_i , inclusive of the contribution of the Gamma prior.

$$\begin{aligned}
Q(\lambda \mid \lambda^*, q^*) &= \sum_{i=1}^K \sum_{j=i+1}^K -(\lambda_i + \lambda_j) \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} \\
&\quad + \sum_{r=1}^R \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K w_{r,ij} \gamma_{r,ij}^* \log \lambda_i \\
&\quad + \sum_{i=1}^K [(a - 1) \log \lambda_i - b \lambda_i] + \text{const.}
\end{aligned}$$

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$$\begin{aligned}
\frac{\partial Q}{\partial \lambda_i} &= - \sum_{j \neq i}^K \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} + \sum_{r=1}^R \frac{\sum_{j \neq i}^K w_{r,ij} \gamma_{r,ij}^*}{\lambda_i} + \frac{a-1}{\lambda_i} - b \\
0 &\stackrel{!}{=} \frac{\sum_{r=1}^R \sum_{j \neq i}^K w_{r,ij} \gamma_{r,ij}^* + (a-1)}{\lambda_i} - \sum_{j \neq i}^K \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} - b \\
\Rightarrow \lambda_i^{(t)} &= \frac{\sum_{r=1}^R \sum_{j \neq i}^K w_{r,ij} \gamma_{r,ij}^{(t-1)} + (a-1)}{\sum_{j \neq i}^K \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^{(t-1)} + w_{r,ji} \gamma_{r,ji}^{(t-1)})}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}} + b} \\
\gamma_{r,ij}^{(t-1)} &= \frac{q_r^{(t-1)} \frac{\lambda_i^{(t-1)}}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}}}{q_r^{(t-1)} \frac{\lambda_i^{(t-1)}}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}} + (1 - q_r^{(t-1)})^{\frac{1}{2}}}
\end{aligned}$$

where the last line furnishes the explicit update at iteration t .

C HYPERPARAMETERS

For Crowd-BT we adopt the implementation of Google (2025) together with its default hyperparameters.

The sole hyperparameters entering the Bayes-BT and BBQ models are the parameters of the prior distributions and the stopping thresholds. For both Bayes-BT and BBQ, we deem the model to have converged once no ELO score varies by more than 1 between successive iterations. The ELO score may be computed from the skill parameter λ according to:

$$\text{ELO} = \log(\text{skill}) \cdot \text{ELO_SCALE_FACTOR}. \quad (14)$$

where the ELO_SCALE_FACTOR is set to 400.

For the skill parameters in both Bayes-BT and BBQ we elected a Gamma prior with shape 5 and rate 0.1; for BBQ, the Beta prior governing rater quality is specified with $\alpha = 10$ and $\beta = 2$.

D DATASETS

We evaluate our models on a diverse collection of human preference datasets spanning both the language and image domains; Table 2 provides an overview.

For natural language we employ the HUMAINE dataset (Team, 2025), a large-scale benchmark comprising in excess of 100k comparisons, alongside MT-Bench (Zheng et al., 2023), which supplies model-comparison judgments elicited from crowd workers on multi-turn dialogues.

For image compression we consider three datasets: WD (Ballé et al., 2025), a dense, expert-labeled dataset furnishing thousands of comparisons per rater; HiFiC (Mentzer et al., 2020), which evaluates learned image codecs; and ConHa (Aczel & Wattenhofer, 2024), which is concerned with conditional generative models. These datasets differ substantially in scale, in the number of raters, and in rater expertise, and together they constitute a broad testbed for robustness.

Finally, we introduce the inhomogeneous rater quality (**IHQ**) dataset, obtained from a user study on the CLIC2024 (CLIC, 2025) data conducted by way of the *Mabyduck* platform. It comprises two-alternative-forced-choice (2AFC) judgments across 28 generative image compression models. In order to study the impact of rater quality, we provide two subsets: (i) a **screened** subset, wherein raters passed pre-screening checks for attention and display quality, and (ii) an **unscreened** subset, wherein all raters are retained. The screened subset is representative of higher-quality raters, whereas the unscreened subset more faithfully reflects the noisy conditions characteristic of large-scale human evaluation.

Table 2: Summary of the datasets employed in our experiments. The IHQ dataset was collected by us on the Mabyduck platform and encompasses both 2AFC and 3AFC settings. For this dataset we provide screened and unscreened subsets: the screened subsets retain only those raters who passed pre-screening tests for attention and display quality, thereby ensuring higher reliability, whereas the unscreened subsets retain all raters.

dataset	# comparisons	# raters	# models	comp/rater	comp/model
HUMAINE (Team, 2025)	105,220	1,977	27	53.2	3897.0
MT-Bench (Zheng et al., 2023)	3,355	65	6	51.6	559.2
WD (Ballé et al., 2025)	16,659	5	30	3331.8	555.3
HiFiC (Mentzer et al., 2020)	5,220	20	9	261.0	580.0
ConHa (Aczel & Wattenhofer, 2024)	1,531	40	8	38.3	191.4
IHQ-all	4,074	112	28	36.4	145.5
IHQ-screened	2,012	50	28	40.2	71.9
IHQ-unscreened	2,062	62	28	33.3	73.6

This assemblage of datasets enables us to study both large-scale, relatively clean benchmarks and smaller, noisier settings in which the modeling of rater reliability is of critical importance.

E USER STUDY PLATFORM

All pairwise comparisons on the CLIC2024 (CLIC, 2025) dataset were collected by means of the Mabyduck platform (Ltd., 2025); a screenshot of the platform is presented in Figure 4. The task poses the question “Which image looks more similar to the reference image?” Ties are disallowed, and all pairs were selected uniformly at random.

In order to study rater quality, we partition the IHQ dataset into screened and unscreened subsets. The screened raters had passed pre-tests for color vision, display contrast, and sensitivity to subtle differences, and had completed training comparisons in which they selected the higher-quality image so as to reinforce the evaluation criteria.

Figure 5 displays four representative images from the pre-screening process. The first two are standard color-blindness tests, in which raters must correctly identify the number shown within each pattern. The final two are shape-detection tests devised to evaluate the raters’ sensitivity to low-contrast objects: the one features a light gray shape upon a white background, the other a dark gray shape upon a black background. Collectively, these pre-screening tests serve to ensure that only raters possessed of adequate visual acuity contribute to the screened subset.

F BOOTSTRAPPING DETAILS

Inasmuch as both BBQ and Crowd-BT model rater quality explicitly, we conduct bootstrapping over raters rather than over individual comparisons. This strategy preserves the comparison distribution of each rater and thereby ensures an equitable assessment of stability. For all datasets we perform 10,000 bootstrap resamples of raters, with the sole exception of the HUMAINE dataset, on which a single bootstrap iteration for Crowd-BT consumes approximately 15 minutes, as discussed in Section 4.4; on HUMAINE we accordingly perform 1,000 resamples so as to curtail the computational burden.

G UNCERTAINTY ESTIMATION DETAILS

Beyond the mere ranking of items, the estimation of uncertainty is of considerable importance for adjudicating whether observed differences are statistically significant.

The Bayesian Bradley–Terry (Bayes-BT) and Bayesian Bradley–Terry with Quality (BBQ) models quantify uncertainty by way of the posterior distribution over item skills. Each skill is furnished with a Gamma prior, which is updated in light of the observed pairwise comparisons. Credible intervals

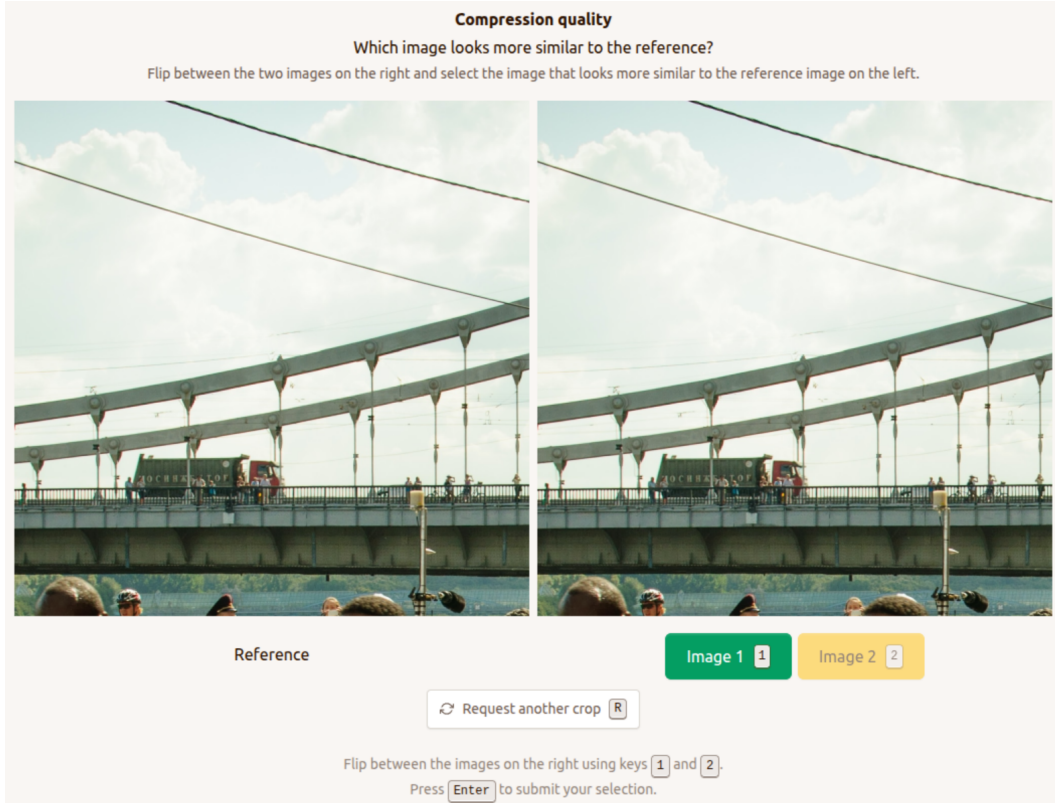


Figure 4: Screenshot of the Mabyduck user study platform employed for the collection of pairwise comparisons. A reference image is displayed on the left, while the rater elects between two compressed images shown on the right.

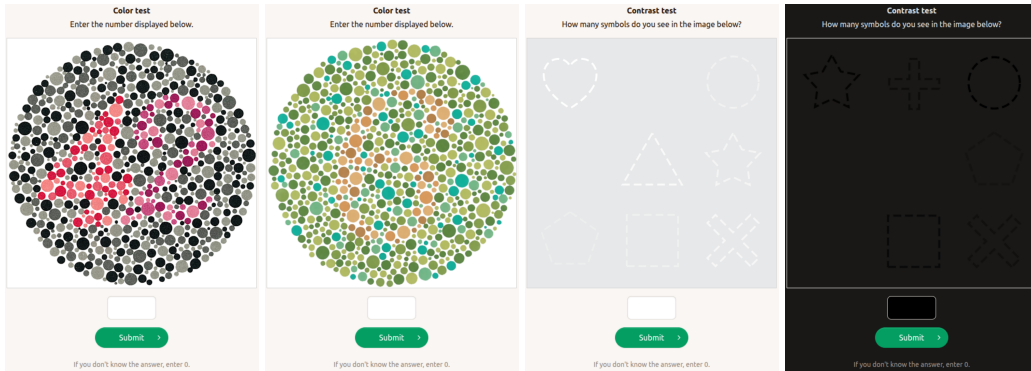


Figure 5: Four representative images drawn from the rater pre-screening process. The first two constitute color-blindness tests, in which raters must identify the number displayed within each pattern. The latter two are shape-detection tests devised to gauge sensitivity to low-contrast objects: one a light gray shape upon a white background, the other a dark gray shape upon a black background.

derived from the posterior are subsequently converted to the Elo scale so as to facilitate comparison across items.

The classical Crowd-BT model, being non-Bayesian, estimates uncertainty by recourse to a frequentist approximation. Specifically, it employs a second-order Taylor expansion about the maximum likelihood estimate. The Hessian of the log-likelihood is inverted so as to obtain the covariance matrix, whose diagonal entries correspond to the variances of the individual items. These variances

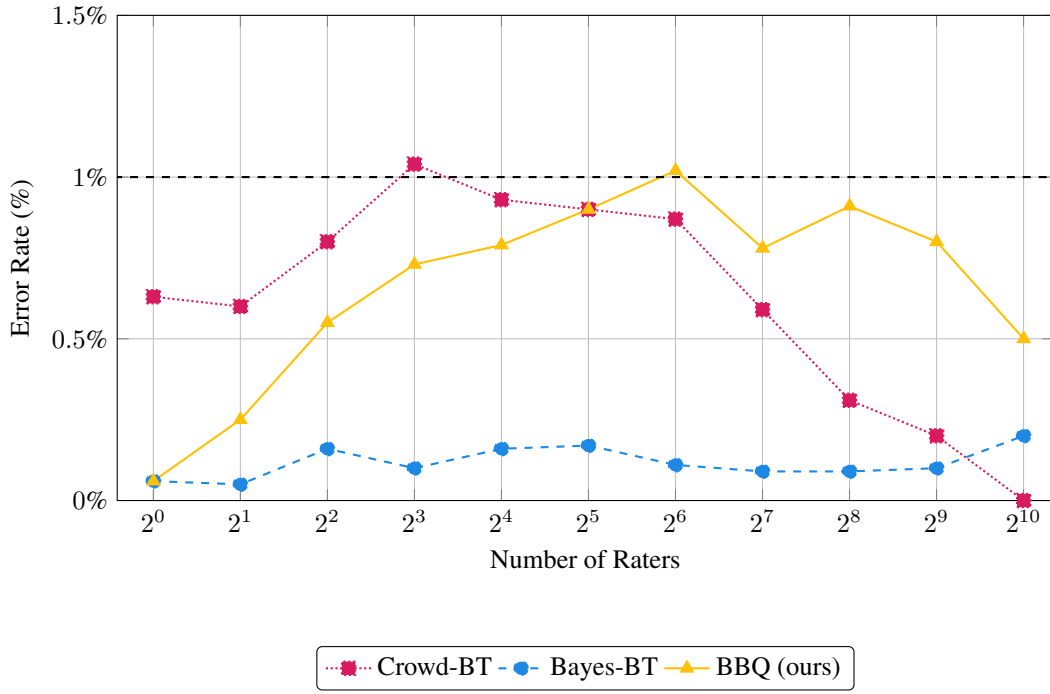


Figure 6: Type I error rate as a function of the number of raters. The dashed line marks the expected error rate of 1%.

are converted into 99% confidence intervals by way of:

$$p_{99} = \sqrt{\text{diag}(\text{covariance})} \times k_{\text{Erfc}0.01} \times \sqrt{2} \approx \sqrt{\text{diag}(\text{covariance})} \times 3.29,$$

where the constants serve to scale the standard deviation so as to match the 99% confidence level. Narrower likelihood peaks give rise to smaller intervals, betokening greater certainty, whereas flatter peaks yield larger intervals, signalling correspondingly greater uncertainty.

To evaluate how faithfully these models estimate uncertainty, we conduct a controlled simulation experiment. We generate trials involving two items of equal skill (a 50–50 win probability) and simulate the pairwise comparison data by means of coin flips. For each trial, multiple raters (users) are simulated, and the models are applied so as to estimate Elo scores together with their associated uncertainty.

Formally, the null hypothesis H_0 asserts that the two items are of equal strength. For each trial we compute the 99% confidence (or credible) interval for the skill estimate of each item. A model is said to reject H_0 erroneously whenever the confidence intervals fail to overlap, thereby signalling a statistically significant difference between items where in truth none exists. The primary metric of interest is thus the frequency with which each model erroneously concludes that the items differ—that is to say, the observed Type I error rate.

We perform 10,000 trials, employing 50 comparisons per rater, while varying the number of raters so as to ascertain how the uncertainty estimates scale with the quantity of data. This design permits us to estimate the empirical Type I error rate for each model and to compare it against the theoretical expectation of 1% at the 99% confidence level. As Figure 6 reveals, Crowd-BT and BBQ exhibit Type I error rates in close proximity to the expected 1%, an indication that their uncertainty estimates are well-calibrated. Bayes-BT, by contrast, proves excessively conservative, consistently producing lower error rates than anticipated—a symptom that its credible intervals are less well-calibrated for the purposes of significance testing.